Ifeanyi Desmond Dike

Senior Software Engineer | Voice AI & Distributed Systems

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PROFILE

Software engineer with **9+ years of experience** building backend services, real-time media and AI products. Owns voice infrastructure from telephony and streaming speech to deployment and evaluation. Recent work spans on-demand call capacity, distributed scheduling and multilingual agents. Strong Python, Go and TypeScript background; built production systems supporting up to 10,000 concurrent connections.

TECHNICAL SKILLS

Languages & backend: Python, asyncio, Go, TypeScript, FastAPI, gRPC, REST, PostgreSQL, Redis, MongoDB, RQ background workers.

Voice & AI: Pipecat, Twilio, WebSockets, WebRTC, streaming STT/TTS, VAD, turn-taking, LLM tool use, RAG, evaluation, PyTorch, Hugging Face.

Infrastructure: Fly.io, Docker, Kubernetes, AWS, GCP, Azure, CI/CD, observability, distributed systems and multi-tenant services.

PROFESSIONAL EXPERIENCE

DataStrut AI

Founding AI Engineer / Senior Software Engineer | Aug 2025 - Present | Remote

- Own the production voice platform across telephony, streaming speech, LLM orchestration and post-call processing; migrated managed voice infrastructure to a self-hosted Python/Pipecat stack.
- Scaled fixed shared capacity into an on-demand Fly Machines pool with per-call isolation, atomic Redis reservations and reserved inbound capacity. Added safe drain/resume lifecycles and fenced callbacks to protect active calls and newer reservations.
- Separated outbound scheduling from report generation with dedicated RQ workers. Added asynchronous, idempotent call triggers and capacity-aware retries to avoid duplicate dials and queue contention.
- Built capacity models and load tests spanning CPU, memory, Redis and report throughput. Validated event ingestion with **6,000 successful synthetic requests** across four load levels; identified Redis connections as the first call-routing constraint.
- Cut steady-state STT session usage by **about 67% (3 to 1 per call)** in three-agent workflows by tying sessions to agent activation. Delivered English/Spanish support across recognition, synthesis, prompts and reporting.
- Built response-admission controls, echo-aware turn-taking and deterministic regression tests; added playback-aligned traces and recordings to distinguish generated speech from what callers heard.

Cenphi

Co-Founder and Lead Developer | Mar 2025 - Present | Remote

- Built an asynchronous video-production pipeline with Python, FastAPI, Whisper, diarization, LLMs and FFmpeg; added job orchestration, provider fallback and live progress updates.
- Developed segment-based story composition and artifact tracking, reducing manual review by **approximately 80%**.

PROFESSIONAL EXPERIENCE - CONTINUED

Fynance Hub

Technical Lead | Oct 2024 - Present | Remote, part-time

- Designed Go/Python microservices with gRPC, owning architecture through production delivery.
- Built PostgreSQL/Redis services for real-time data processing.
- Developed GPT-4 and RAG-based recommendations for broker lead management.
- Automated lead verification through outbound voice calls.

Arbor Platform Inc.

Senior Software Engineer | Feb 2023 - Oct 2024 | Remote

- Built Claude/OpenAI-powered video workflows that reduced production time by **60%** and improved throughput by **80%**.
- Delivered text-to-video and audio-transcription features for AI-assisted editing.
- Built a content studio and B-roll asset-management tools with one-click social distribution.
- Developed React/TypeScript interfaces and integrated backend APIs and cloud services.

Howdy Learning

Senior Software Development Engineer | May 2020 - Mar 2023 | Remote

- Reduced WebSocket network traffic by **more than 99%** through real-time communication optimizations.
- Built live-video infrastructure supporting up to **10,000 concurrent connections**.
- Developed a GetStream/WebSocket chat system serving approximately **100 communities**.
- Delivered embedded video, voice and SSO widgets for Thinkific and WordPress, with integrations used by **more than 100 customers**.

Bujuwa Nigeria Limited

Software Developer | Jul 2018 - Dec 2019 | Abuja, Nigeria

- Delivered web and mobile assessment systems, from requirements and architecture to implementation.

OPEN SOURCE & SELECTED ENGINEERING WORK

Pipecat: Reported Anthropic stale-generation failures with reproduction evidence in #5035 and #5036; #5036 was resolved upstream in PR #5041.

Acoustic Turn-over Rate (ATR): Built ASR-independent evaluation: three VAD backends agreed within 7.6 percentage points, versus up to 23.5 points of ASR disagreement on the same audio. Code | Research artifact.

Full-Duplex-Bench: Exposed up to 62x metric variation across ASR backends on identical audio. Code and results.

Cenphi: Co-founded an AI video platform for transcript-linked story composition, speaker-aware analysis and automated editing. cenphi.io.

EDUCATION

M.Sc., Financial Engineering | WorldQuant University | 2026 - Present (in progress)

B.Tech., Applied Mathematics, First Class | Federal University of Technology, Akure | 2018
GPA: 4.63/5.00. National university mathematics competition: Gold (2017), Bronze (2016).